

Each question is worth 5 points for a total of 40 points. The extra Credit can get you an additional 5 points making the highest possible score 45 out of 40. Write solutions using COMPLETE sentences.

Problem 1

For each of the following relations on the set $\{1, 2, 3, 4, 5\}$, determine whether the relation is reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

- a) $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5)\}$
- b) $R = \{(1, 2), (2, 3), (3, 4), (4, 5)\}$
- c) $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5)\}$
- d) $R = \{(1, 1), (1, 2), (2, 1), (3, 4), (4, 3)\}$
- e) $R = \{1, 2, 3, 4, 5\} \times \{1, 2, 3, 4, 5\}$

Problem 2

Prove that if x and y are both *odd*, then $x \equiv y \pmod{2}$.

Prove that if x and y are both even, then $x \equiv y \pmod{2}$.

Problem 3

How many different anagrams (including nonsensical words) can be made from SUCCESS if we require that the first and last letters must both be S.

Problem 4

Write out all the 3-element subsets of $\{1, 2, 3, 4, 5, 6\}$ to verify that $\binom{6}{3} = 20$

Problem 5

Twenty people attend a party. If everyone shakes everyone else's hand exactly once, how many handshakes take place?

Problem 6

Which of the following relations R defined on the set of all human beings (not just those in your family) are equivalence relations?

- a) xRy provided x and y have the same mother.
- b) xRy provided x and y have the same mother and the same father.
- c) xRy provided x and y have at least one parent in common.

Problem 7

Let $R = \{(x, y) : x, y \in \mathbb{Z} \text{ and } |x| = |y|\}$

a) Prove that R is an equivalence relation on the integers.

b) Find the equivalence classes $[5]$, $[-2]$, and $[0]$.

Problem 8

The expression $(x + 2)^{50}$ is expanded. What is the coefficient of x^{17} ?

Extra Credit:

Let n and k be positive integers with $k + 2 \leq n$. Prove the identity

$$\binom{n+2}{k+2} = \binom{n}{k} + 2\binom{n}{k+1} + \binom{n}{k+2}.$$